

»» Chapter XVI ««

THE MODULAR FUNCTION λ . THE PICARD THEOREMS

PREFATORY REMARKS

The little theorem of Picard says that if $v_f(w) = 0$ for three distinct $w \in \hat{\mathbb{C}}$, where f is meromorphic on $\mathbb{C}$, then f is constant. The big theorem of Picard says that if the same condition holds for f meromorphic on $\Delta(a; r) - \{a\}$, then f is the restriction of a function meromorphic on $\Delta(a; r)$. In this chapter we follow the old classical route of Picard to these famous theorems, proceeding via the modular function λ to be introduced below. To be sure, the prospective specialist in complex function theory will want to go far beyond Picard's original program and learn the beautiful Nevanlinna theory of meromorphic functions, which has roots in the Picard theorems, and its modern developments. The account of W. K. Hayman [51] renders the Nevanlinna theory accessible to the English-reading student.

Apart from putting the subsequent study of the Nevanlinna theory in perspective and playing a motivating role, the approach we use recommends itself both because it brings into play notions from the earlier chapters of our discipline—such as that of the p -function—which serve as a starting point for the study of modular functions (thereby revealing a wealth of mathematical ideas implicit in the classical developments of complex function theory) and also because it introduces important examples from the theory of automorphic groups and functions—the modular group MG and a modular function. The lively renewed interest in this theory justifies encounter with such examples, embryonic though they may be.

Another advantage of the old road is that it furthers the student's experience with covering maps and liftings, the general mathematical importance of which needs no apology. He has already met such notions in dealing with the exponential function.

1. THE MODULAR FUNCTION λ

We shall single out for study an analytic function, to be denoted λ , with domain $\{\text{Im } z > 0\}$, which emerges in the course of examining the dependence of the pe-function on the entering parameters. It will be seen to enjoy interesting transformation properties relative to the modular group of Ex. 1.6, Ch. XV, which we shall denote MG . We shall be principally concerned in our study of λ with showing:

The set of functions f analytic on $\Delta(0; r)$ and mapping $\Delta(0; r)$ into $\mathbb{C} - \{0, 1\}$ are the functions of the form $\lambda \circ g$ where g is analytic on $\Delta(0; r)$ and maps $\Delta(0; r)$ into $\{\text{Im } z > 0\}$.

This property of λ is the exact counterpart of the property enjoyed by $\exp$ relative to the functions analytic on $\Delta(0; r)$ which map $\Delta(0; r)$ into $\mathbb{C} - \{0\}$.† We may already anticipate the small Picard theorem, setting $r = +\infty$ and applying the theorem of Liouville to $1/(g + i)$. The general situation is reduced to the case where $0, 1, \infty$ are omitted with the aid of a Möbius transformation mapping a set of three distinct omitted values onto $\{0, 1, \infty\}$.

We define the *modular function* λ as the function with domain $\{\text{Im } z > 0\}$ given by

$$\lambda(\alpha) = \frac{p[(1 + \alpha)/2] - p(1/2)}{p(\alpha/2) - p(1/2)}, \quad (1.1)$$

where p is the Weierstrass pe-function associated with the additive subgroup of $\mathbb{C}$ generated by $\{1, \alpha\}$. It is to be recalled (Exs. 2.2 and 2.5, Ch. XIV) that p attains each point of $\hat{\mathbb{C}}$ twice on $\{s + t\alpha: 0 \leq s, t < 1\}$, the count being by multiplicity, and that p' takes the value 0 at each of the points: $1/2, \alpha/2, (1 + \alpha)/2$. Hence p takes distinct finite values at these three points. We see from (1.1) that $\lambda(\alpha) \neq 0, 1$ and by the definition of p that λ is analytic on $\{\text{Im } z > 0\}$.‡

To continue we shall want the first result of the following exercise.

† The factorization in terms of composition, $f = \lambda \circ g$, yields g as a “lifting” of f . λ is a universal covering map of $\{\text{Im } z > 0\}$ onto $\mathbb{C} - \{0, 1\}$.

‡ It is to be remarked that a certain amount of harmless freedom prevails in the normalization of λ and that some variation will be detected in the literature. We refer to the possibility of forming ratios $(C - A)/(B - A)$ with $\{A, B, C\} = \{p(1/2), p(\alpha/2), p[(1 + \alpha)/2]\}$ in different ways.

EXERCISE

1.1. Show by studying the principal part of $(p')^2$ at 0 that $(p')^2 = Q \circ p$ where $Q \in M(\hat{\mathbb{C}})$ has a pole of order 3 at ∞ and elsewhere is analytic (that is, $Q|C$ is a polynomial of degree 3 and $Q(\infty) = \infty$). Determine the coefficients of Q in terms of α .

Suppose that u is meromorphic on some disk $\Delta(0; r)$, has a pole at 0, and satisfies $(u')^2 = c^2 Q \circ u$ where Q is the function of Ex. 1.1 and $c \in \mathbb{C} - \{0\}$. By examining the 0-Laurent coefficients of u , we see that u has a pole of order 2 at 0 and that the coefficient of index -2 is c^{-2} . If similarly v is meromorphic on $\Delta(0; \rho)$ and has a pole at 0 and satisfies $(v')^2 = d^2 Q \circ v$, $d \in \mathbb{C} - \{0\}$, we clearly have $v(z) = u[\varphi(z)]$ for z small where φ is analytic at 0 and satisfies $\varphi(0) = 0$, $\varphi'(0) \neq 0$. On referring to the differential equations satisfied by u and v we conclude that $[\varphi'(z)]^2 = (d/c)^2$ for z small and hence that either $\varphi(z) = (d/c)z$, z small, or $\varphi(z) = -(d/c)z$, z small.

We shall use the relation we have just established to show that λ is *not constant*. In order to bring to the fore the parameter involved we shall denote by $p^{[\alpha]}$ the pe-function associated with the group generated by $\{1, \alpha\}$.† Suppose that $\lambda(\beta) = \lambda(\alpha)$. Then there exist complex constants $A (\neq 0)$, B such that $A^2 p^{[\beta]} + B$ attains the same values with multiplicity 2 as does $p^{[\alpha]}$. Since the zeros of Q are the finite values attained by $p^{[\alpha]}$ with multiplicity 2, we conclude that $A^2 p^{[\beta]} + B$ satisfies $(u')^2 = A^{-2} Q \circ u$. Hence $A^2 p^{[\beta]}(z) + B = p^{[\alpha]}(z/A)$ and we see that $1/A$ and β/A are in the group generated by $\{1, \alpha\}$ and that A and αA are in the group generated by $\{1, \beta\}$. It follows that there exists $T \in MG$ such that $\beta = T(\alpha)$. The fact that λ is not constant follows from the countability of MG , but it may be established also as follows. If

$$T(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{Z}, \quad ad - bc = 1,$$

then

$$\operatorname{Im}[T(z)] = \frac{\operatorname{Im} z}{|cz + d|^2}, \quad cz + d \neq 0.$$

Hence

$$\operatorname{Im}[T(z)] \leq \max\{\operatorname{Im} z, 1/\operatorname{Im} z\}, \quad \operatorname{Im} z > 0. \quad (1.2)$$

Thus $\lambda(\beta) \neq \lambda(\alpha)$ when $\operatorname{Im} \beta > \max\{\operatorname{Im} \alpha, 1/\operatorname{Im} \alpha\}$.

† In the course of the exposition we shall use the notation $P(z; \omega_1, \omega_2)$ for the value at z of the pe-function associated with the group generated by $\{\omega_1, \omega_2\}$.

Let us now consider the transformation properties of λ . From (1.1) we conclude

$$\lambda(\alpha + 1) = [\lambda(\alpha)]^{-1} \quad (1.3)$$

and

$$\lambda(-1/\alpha) = 1 - \lambda(\alpha), \quad (1.4)$$

$\text{Im } \alpha > 0$.

We consider the values taken by λ when $\text{Re } \alpha = 0$. Using the notation introduced two paragraphs back we see that

$$P(iz; i, i\alpha) = -P(z; 1, \alpha)$$

and hence that $p^{[\alpha]}(z)$ is ∞ or real when z is pure imaginary. Since $p^{[\alpha]}(1 - z) = p^{[\alpha]}(z)$, we see that $p^{[\alpha]}(2^{-1} + it)$ is even, that is, it is equal to $p^{[\alpha]}(2^{-1} - it)$, and when t is real, $p^{[\alpha]}(2^{-1} - it)$ is the conjugate of $p^{[\alpha]}(2^{-1} + it)$. Hence $p^{[\alpha]}(2^{-1} + it)$ is real when t is real. We conclude: $\text{Im } \lambda(\alpha) = 0$, when $\text{Re } \alpha = 0$. Referring to (1.3) we see: $\text{Im } \lambda(\alpha) = 0$, when $|\text{Re } \alpha| = 1$. Applying (1.4) with $\text{Re } \alpha = -1$ we conclude: $\text{Im } \lambda[(1 + e^{i\theta})/2] = 0$, $0 < \theta < \pi$.

Suppose that $S \in MG$. We are interested in the action of S on λ . Let $\beta = S(\alpha)$. The additive subgroup of $\mathbb{C}$ generated by $\{1, \beta\}$ is the subgroup generated by $\{\rho, \rho\alpha\}$ for some $\rho \in \mathbb{C} - \{0\}$. We see, using the differential equation of Ex. 1.1, this section, for the pe-function, that the sets

$$\{P(1/2; 1, \beta), P(\beta/2; 1, \beta), P[(1 + \beta)/2; 1, \beta]\}$$

and

$$\{\rho^{-2} P(1/2; 1, \alpha), \rho^{-2} P(\alpha/2; 1, \alpha), \rho^{-2} P[(1 + \alpha)/2; 1, \alpha]\}$$

are equal. We conclude on examining the ratios formed from these quantities that

$$\lambda(\beta) \in \{T_k[\lambda(\alpha)]: k = 1, \dots, 6\}$$

where $T_1(z) = z$, $T_2(z) = 1/z$, $T_3(z) = 1 - z$, $T_4 = T_2 \circ T_3$, $T_5 = T_2 \circ T_3 \circ T_2$, $T_6 = T_3 \circ T_2$. It is readily seen that $T_1, \dots, T_6$ constitute the group G_0 of Möbius transformations which map $\{0, 1, \infty\}$ onto itself and that $T_j \neq T_k$, $j \neq k$. Since λ is not constant, we may take α_0 , $\text{Im } \alpha_0 > 0$, such that $T_j[\lambda(\alpha_0)] \neq T_k[\lambda(\alpha_0)]$, $j \neq k$. It follows from the continuity of $\lambda \circ S$ and $T_k \circ \lambda$ that $\lambda[S(\alpha)] = T_k[\lambda(\alpha)]$ for a unique k when α is near α_0 . We conclude that $\lambda \circ S = T(S) \circ \lambda$ where $T(S)$ is a unique member of G_0 . It is clear that the map assigning $T(S)$ to S is a homomorphism of MG onto G_0 as we see with the aid of (1.3) and (1.4).

EXERCISE

1.2. Show that there exists a nonconstant function f meromorphic on $\{\operatorname{Im} z > 0\}$ and satisfying $f \circ S = f$, $S \in MG$. Hint: Introduce $g \in M(\hat{\mathbb{C}})$ nonconstant and satisfying $g \circ T = g$, $T \in G_0$.

Show that f meromorphic (resp. analytic) on $\{\operatorname{Im} z > 0\}$ satisfying $f \circ S = f$, $S \in MG$, are of the form $h \circ \lambda$ where h is meromorphic (resp. analytic) on the image of λ . [We shall see presently that the image of λ is $\mathbb{C} - \{0, 1\}$.] Obtain a necessary and sufficient condition on such h for $h \circ \lambda$ to satisfy $h \circ \lambda \circ S = h \circ \lambda$, $S \in MG$.

We turn next to the study of the mapping properties of λ on the set E defined as the union of

$$E_1 = \{x + yi : x \in]-1, 0], y \in \mathbf{P}, |x + yi + 2^{-1}| > \tfrac{1}{2}\},$$

and

$$E_2 = \{x + yi : x \in]0, 1], y \in \mathbf{P}, |x + yi - 2^{-1}| \geq \tfrac{1}{2}\}.\dagger$$

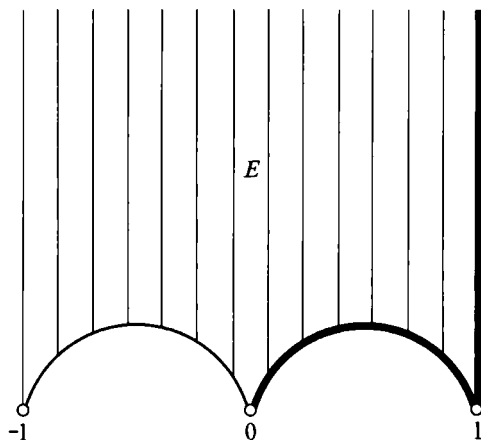


FIG. 16.1

We introduce v_E , the E -valence of λ , which differs from valence as defined earlier in that only points of E are regarded in calculating its value at a point of $\hat{\mathbb{C}}$. That is

$$v_E(w) = \sum_{\lambda|E(z)=w} n(z; \lambda).$$

$\dagger$ We recall that $\mathbf{P}$ denotes the set of positive real numbers.

We shall now show that

$$v_E(w) = 1, \quad w \in \mathbb{C} - \{0, 1\}.$$

To establish this assertion, we note that thanks to (1.2) the function φ_∞ analytic on $\Delta(0; 1) - \{0\}$ satisfying

$$\varphi_\infty(e^{\pi i \alpha}) = \lambda(\alpha) \quad (1.5)$$

has a pole or removable singularity at 0. Indeed, by (1.2) we see that $\lambda(\alpha) \notin \lambda[\Delta(i/2; 1/4)]$ when $\text{Im } \alpha \geq 4$, whence the asserted property of φ_∞ . We let L_∞ denote $\lim_0 \varphi_\infty$. We introduce similarly φ_0 satisfying $\varphi_0(e^{\pi i \alpha}) = \lambda(-\alpha^{-1})$ and let $L_0 = \lim_0 \varphi_0$, and finally φ_1 satisfying $\varphi_1(e^{\pi i \alpha}) = \lambda(1 - \alpha^{-1})$ and let $L_1 = \lim_0 \varphi_1$. We now show that v_E is finite-valued and continuous on $\hat{\mathbb{C}} - \{L_0, L_1, L_\infty\}$. The finiteness of $v_E(w)$, $w \neq L_0, L_1, L_\infty$, follows from the behavior of λ in E near $-1, 0, 1, \infty$, in the neighborhood of which points $\lambda|_E$ takes values near one of L_0, L_1, L_∞ . If v_E were not continuous at $w \neq L_0, L_1, L_\infty$, there would exist (w_n) , $\lim_{n \rightarrow \infty} w_n = w$, $v(w_n) > v(w)$ for all $n \in \mathbb{N}$. We may suppose that $w_n = \lambda(z_n)$, $z_n \in E$, and that $\lim_{n \rightarrow \infty} z_n = z \in \bar{E}$, $\text{Im } z > 0$, and $\lambda(z) \neq w$. This follows on considering the set $\mathcal{E}$ of points z lying in $\bar{E}$, $\text{Im } z > 0$, at which $\lambda(z) = w$ and noting that the z_n may be chosen to lie outside of an open set $O \supset \mathcal{E}$ such that on $O \cap E$ the function λ attains no value more than $v_E(w)$ times counted by multiplicity. The contradiction is manifest. We conclude that v_E takes the same positive integer value at each point of $\hat{\mathbb{C}} - \{L_0, L_1, L_\infty\}$. Hence since v_E takes the value 0 at 0, 1, and ∞ , $\{0, 1, \infty\} \subset \{L_0, L_1, L_\infty\}$. It follows that $\{0, 1, \infty\} = \{L_0, L_1, L_\infty\}$.

There remains to be shown that v_E takes the value 1 on $\mathbb{C} - \{0, 1\}$. To that end we note that when $\alpha \in E$ has a large imaginary part and $\beta \in E$ satisfies $\lambda(\beta) = \lambda(\alpha)$, then $\text{Im } \beta > 1$ since L_0, L_1, L_∞ are distinct and by (1.2) we have

$$\text{Im } \alpha \leq \max\{\text{Im } \beta, 1/\text{Im } \beta\}.$$

Hence for such α , if $\beta \neq \alpha$, we have $\beta = \alpha - 1$ or $\beta = \alpha + 1$; for otherwise $\text{Im } \beta \leq 1/\text{Im } \alpha$. Using (1.3) we find $\lambda(\alpha) = -1$. This is not possible when $\text{Im } \alpha$ is large. Our assertion follows, since we are assured that λ' is not the constant 0. We shall see that $\lambda'(\alpha) \neq 0$, $\text{Im } \alpha > 0$, and even more.

EXERCISES

1.3. Show with the aid of (1.3) and (1.4) that $L_\infty = 1$, $L_0 = 0$, $L_1 = \infty$.

1.4. Show that $\lambda(\text{int } E_1) = \{\text{Im } z > 0\}$ and $\lambda(\text{int } E_2) = \{\text{Im } z < 0\}$. [Reference is made to the sets introduced in the definition of E .]

Our principal goal is to show that λ has the following important property:
Given $a \in \mathbb{C} - \{0, 1\}$ there exists r positive such that $\lambda|_C$ is a homeomorphism

of C onto $\Delta(a; r)$ for each component C of $\lambda^{-1}[\Delta(a; r)]$.[†] To establish this assertion, suppose that $a = \lambda(\beta)$. There exists $\alpha \in E$ such that $a = \lambda(\alpha)$. Hence $\beta = S(\alpha)$, $S \in MG$. We take ρ , $0 < \rho < \text{Im } \alpha$, so small that $\lambda|_{\Delta(\alpha; \rho)}$ is univalent and $\lambda[\Delta(\alpha; \rho)] \cap T \circ \lambda[\Delta(\alpha; \rho)] = \emptyset$ for $T \in G_0$ satisfying $T(a) \neq a$. Using the relation $\lambda \circ S = T(S) \circ \lambda$, $T(S) \in G_0$, established above, we see that $\lambda'(\beta) \neq 0$ and that $\lambda|_{S[\Delta(\alpha; \rho)]}$ is univalent and has as image $T(S)\{\lambda[\Delta(\alpha; \rho)]\}$, a region containing a . If we take r so small that $\Delta(a; r) \subset T\{\lambda[\Delta(\alpha; \rho)]\}$ for each $T \in G_0$ such that $a = T(a)$, then each component C of $\lambda^{-1}[\Delta(a; r)]$ is mapped univalently by λ onto $\Delta(a; r)$. For $C \cap S[\Delta(\alpha; \rho)] \neq \emptyset$ for some $S \in MG$ satisfying $T(S)(a) = a$ and hence C is the image of $\Delta(a; r)$ with respect to the inverse of $\lambda|_{S[\Delta(\alpha; \rho)]}$ since C contains the image in question but cannot contain it properly (for otherwise it would contain a frontier point of the image which would be taken by λ into a point of $C(a; r)$). The covering property of λ is established.

The fact that λ is a covering map leads to its lifting property. We first show:

Let f be a continuous map of $[a, b]$ into $\mathbb{C} - \{0, 1\}$, $-\infty < a < b < +\infty$. Let $c \in [a, b]$ and suppose that $f(c) = \lambda(\beta)$. Then there exists a unique continuous map g of $[a, b]$ into $\{\text{Im } z > 0\}$ satisfying $f = \lambda \circ g$, $g(c) = \beta$.

PROOF: *Uniqueness.* This is a consequence of the observation that if g and h are continuous maps of a connected topological space X into $\{\text{Im } z > 0\}$ satisfying $\lambda \circ g = \lambda \circ h$, then either $g = h$ or $g(p) \neq h(p)$, $p \in X$. Indeed, by the continuity of the maps and the Hausdorff property of $\mathbb{C}$ the set $\{g(p) \neq h(p)\}$ is open. If q is a point of $\{g(p) = h(p)\}$, by the univalence of λ restricted to a sufficiently small neighborhood of $g(q)$, $\lambda[g(p)] = \lambda[h(p)]$ implies $g(p) = h(p)$ for p near q . Hence $\{g(p) = h(p)\}$ is open. The asserted relation between g and h follows from the connectedness of X . It is to be noticed that the full force of the fact that f is a covering map was not invoked here, only the local univalence of λ .[‡]

Existence. We consider a finite covering of the image of f by disks $\delta_k = \Delta(z_k; r_k)$, $k = 1, \dots, m$, such that λ maps each component of $\lambda^{-1}(\delta_k)$ univalently onto δ_k . If $n \in \mathbb{N}$ is sufficiently large, then for each $j \in \langle 0, 2^n - 1 \rangle$ there exists $k(j) \in \langle 1, m \rangle$ such that

$$f([t_j, t_{j+1}]) \subset \delta_{k(j)}$$

where $t_j = a + 2^{-n}j(b - a)$. The proof of this assertion by the method of successive bisections is routine. Let $c \in [t_v, t_{v+1}]$ and let μ_v denote the inverse of λ

[†] This says that λ is a *covering map* of $\{\text{Im } z > 0\}$ onto $\mathbb{C} - \{0, 1\}$ in the sense of topology.

[‡] A reference to the uniqueness of the continuation of an element along a path is appropriate here.

restricted to the component of $\lambda^{-1}[\delta_{k(v)}]$ containing β . We propose $g(t)$ as $\mu_v[f(t)]$, $t_v \leq t \leq t_{v+1}$. Thereupon (when $n > 0$) we proceed to adjoining segments. Thus if $v > 0$, we introduce μ_{v-1} as the inverse of λ restricted to the component of $\lambda^{-1}[\delta_{k(v-1)}]$ containing $\mu_v[f(t_{v-1})]$ and propose $g(t)$ as $\mu_{v-1}[f(t)]$, $t_{v-1} \leq t \leq t_v$. We proceed similarly for $[t_{v+1}, t_{v+2}]$ when $v < 2^n$. The construction is continued in this manner and the function g so obtained fulfills the imposed conditions.

For the case of a continuous map f of a rectangle $[a, b; c, d]$ into $\mathbf{C} - \{0, 1\}$ we proceed along similar lines starting with subrectangles obtained by successive quadrisections which are so fine that the image of each with respect to f lies wholly in a δ_k as above. We take the subrectangle containing the point of normalization $s + ti$: $f(s + ti) = \lambda(\beta)$, and construct $g(z)$ as above. A divergence from the 1-dimensional case arises when we proceed to adjacent subrectangles, for two adjacent subrectangles have a segment rather than a point in common. Since a segment is connected, we see that the g for two adjacent subrectangles adjusted to take the same value at a point of the segment common to the two subrectangles take the same value at each point of the segment. We proceed first along the horizontal "row" which has the preferred subrectangle containing (s, t) and thereupon proceed in the vertical direction along each "column" starting from the subrectangle of the preferred "row." The connectedness of the intersections of adjacent vertical strips assures coherence and we obtain g continuous on $[a, b; c, d]$ having the property that $f = \lambda \circ g$ and $g(s + ti) = \beta$.

We see by the reasoning employed in the study of continuous logarithms that continuous maps f of $\{\text{Im } z > 0\}$ or $\mathbf{C}$ into $\mathbf{C} - \{0, 1\}$ admit normalized liftings g as above. For the case where the domain is $\{\text{Im } z > 0\}$ we proceed, when the point where f is normalized is c , by taking the union of the normalized liftings of

$$f|[-(n+1) + \text{Re } c, (n+1) + \text{Re } c; (\text{Im } c)/(n+1), (n+1)(\text{Im } c)],$$

$n \in \mathbf{N}$. For a continuous map f of a disk $\Delta(a; r)$ into $\mathbf{C} - \{0, 1\}$ we may reduce the problem to one where the domain is an upper half plane by the intervention of

$$a + r \frac{z - i}{z + i}, \quad \text{Im } z > 0,$$

when a and r are finite and an obvious variant when $a = \infty$ and r is finite. In the case where the domain is $\mathbf{C}$ our assertion is now immediate. For the stated domains it is easy to see that if f is analytic on its domain, then so is g . Indeed, if g is a continuous map of a region $A \subset \hat{\mathbf{C}}$ into $\{\text{Im } z > 0\}$ such that $\lambda \circ g$ is analytic on A , then g is analytic on A . For given $a \in A$, let δ denote a disk in $\{\text{Im } z > 0\}$ centered at $g(a)$ so small that $\lambda|_\delta$ is univalent. We take

$\Delta(a; r) \subset A$ so that $g[\Delta(a; r)] \subset \delta$ and let μ denote the inverse of $\lambda|_{\delta}$. Then $\mu \circ [(\lambda \circ g)|_{\Delta(a; r)}]$ is analytic on $\Delta(a; r)$ and is equal to $g|_{\Delta(a; r)}$.

The small Picard theorem is now under control.

It is to be emphasized that our object in introducing λ is precisely to have a function analytic on $\{\text{Im } z > 0\}$ which is a covering map in the indicated sense of $\{\text{Im } z > 0\}$ onto $\mathbb{C} - \{0, 1\}$. The wealth of interesting properties of λ which emerge is a bonus.

EXERCISES

1.5. If λ_1 is analytic on $\{\text{Im } z > 0\}$ and is a covering map of $\{\text{Im } z > 0\}$ onto $\mathbb{C} - \{0, 1\}$ in the sense specified in the text, then $\lambda_1 = \lambda \circ S$ where S is a conformal automorphism of $\{\text{Im } z > 0\}$.

1.6. If S is a continuous map of $\{\text{Im } z > 0\}$ into itself such that $\lambda = \lambda \circ S$, then $S \in MG$.

1.7. Given f analytic on $\Delta(0; 1)$ with image in $\mathbb{C} - \{0, 1\}$. Let

$$g(z) = \sum_0^{\infty} a_k z^k, \quad |z| < \rho,$$

where $\rho > 0$, and let $f(z) = \lambda[g(z)]$ for z small. Using the fact that λ is a covering map of $\{\text{Im } z > 0\}$ onto $\mathbb{C} - \{0, 1\}$ show that $\rho \geq 1$, that $\text{Im } g(z) > 0$, and that $f(z) = \lambda[g(z)]$, $|z| < 1$. Appeal is not to be made to the existence of a normalized analytic lifting with respect to λ . Hint: Suppose that $\rho < 1$ and show using the covering map property of λ that $\text{Im } g(z) > 0$, $|z| < \rho$, and that, for each $u \in C(0; 1)$, $\lim_{\rho} g(tu)$ exists and is in $\{\text{Im } z > 0\}$. Here $g(tu)$ has domain $[0, \rho]$. For each such u let h_u denote a function analytic on a disk with center ρu such that $\lambda[h_u(z)] = f(z)$, z on the domain of h_u (assumed in $\Delta(0; 1)$) and satisfying $h_u(\rho u) = \lim_{\rho} g(tu)$. Thereupon consider

$$g \cup \bigcup_{|u|=1} h_u,$$

and complete the proof noting that $\text{Im } g(z) > 0$, $|z| < 1$.

It is to be remarked that we could use the result of this exercise rather than the more general lifting result which was obtained for continuous f . Preference was given to the reasoning of the text because of its more general character.

1.8. Let A be a homotopically simply connected region $\subset \hat{\mathbb{C}}$. Let f be a continuous map of A into $\mathbb{C} - \{0, 1\}$. Show that if $f(a) = \lambda(\alpha)$, where a is a given point of A , there exists a continuous map g of A into $\{\text{Im } z > 0\}$ such that $f = \lambda \circ g$, $g(a) = \alpha$. [Of course, g is analytic on A when f is, as follows from the remarks of the text.] Hint: Let Φ denote the family of paths γ in A with domain $[0, 1]$ satisfying $\gamma(0) = a$. Let G_{γ} denote the unique continuous map of $[0, 1]$ into $\{\text{Im } z > 0\}$ satisfying: $G_{\gamma}(0) = \alpha$, $\lambda \circ G_{\gamma} = f \circ \gamma$. Show that

$$\{(\gamma(1), G_{\gamma}(1)) : \gamma \in \Phi\}$$

serves for g .

2. THE BIG PICARD THEOREM

As a preliminary to the proof of the big Picard theorem we establish the following theorem of a type due to Schottky.†

Theorem 2.1: *Let $0 < \delta < 1$. Let Φ denote the class of functions analytic on $\Delta(0; 1)$ with image in $\mathbb{C} - \{0, 1\}$ and such that $\chi[f(0), a] \geq \delta$, $a = 0, 1, \infty$. Let*

$$m(r) = \inf\{\chi[f(z), a]: a = 0, 1, \infty, f \in \Phi, |z| \leq r\}.$$

Then $m(r) > 0$, $0 \leq r < 1$.

PROOF: We put aside the trivial cases where the Φ are empty. Given $f \in \Phi$, let $\alpha(f)$ satisfy: $\lambda[\alpha(f)] = f(0)$, $\alpha(f) \in E$, where E is the set so designated in §1, this chapter. Let c denote the supremum of the set of positive numbers

$$\max\{\operatorname{Im} \alpha(f), [\operatorname{Im} \alpha(f)]^{-1}\},$$

all $f \in \Phi$ being taken into account. It follows from the behavior of $\lambda|_E$ that $c < +\infty$. Given y , $1 \leq y < +\infty$, let $\mu(y)$ denote the chordal distance from

$$\lambda(\{y^{-1} \leq \operatorname{Im} z \leq y\})$$

to $\{0, 1, \infty\}$. Now let g be analytic on $\Delta(0; 1)$ and satisfy: $\lambda \circ g = f$, $g(0) = \alpha(f)$. Using the Harnack inequality (1.7), Ch. XIII, we obtain

$$[\operatorname{Im} \alpha(f)] \frac{1 - |z|}{1 + |z|} \leq \operatorname{Im} g(z) \leq [\operatorname{Im} \alpha(f)] \frac{1 + |z|}{1 - |z|}, \quad |z| < 1,$$

so that

$$\chi[f(z), a] \geq \mu\left(c \frac{1 + |z|}{1 - |z|}\right),$$

$|z| < 1$, $a = 0, 1, \infty$. The theorem follows.

The big Picard theorem. It is immediate that the big Picard theorem stated at the outset of this chapter may be reduced to the following theorem.

Theorem 2.2: *Let f be analytic on $\{1 < |z| < +\infty\}$ and have image in $\mathbb{C} - \{0, 1\}$. Then f does not have an essential singularity at ∞ .*

Indeed, if f satisfies the hypotheses of the big Picard theorem and has image in $\hat{\mathbb{C}} - \{A, B, C\}$ and r is finite (as we may suppose), then it suffices

† The notation χ below refers to the chordal distance (cf. §4, Ch. VI).

to consider $M[f(a + z^{-1}r)]$ when $a \in \mathbb{C}$ and $M[f(r^{-1}z)]$ when $a = \infty$, $1 < |z| < +\infty$, where M is the Möbius transformation satisfying $M(A) = 0$, $M(B) = 1$, $M(C) = \infty$.

PROOF OF THEOREM 2.2: We distinguish two cases. (1) For some $\rho \in]1, +\infty[$, we have

$$f[C(0; r)] \cap C(0; 2) = \emptyset,$$

$\rho < r < +\infty$. Since $f(\{\rho < |z| < +\infty\})$ is connected and does not intersect $C(0; 2)$, it is contained in $\Delta(0; 2)$ or in $\Delta(\infty; \frac{1}{2})$. When

$$f(\{\rho < |z| < +\infty\}) \subset \Delta(0; 2),$$

f has a removable singularity at ∞ . When

$$f(\{\rho < |z| < +\infty\}) \subset \Delta(\infty; \tfrac{1}{2}),$$

f has a removable singularity or a pole at ∞ . (2) The assumption of case (1) does not hold. Let $F(z) = f(e^z)$, $\operatorname{Re} z > 0$. Clearly, F is periodic with period $2\pi i$. There exists $(z_n)_{n \in \mathbb{N}}$ satisfying: (a_n) is increasing, $a_0 > 2\pi$, $\lim_{n \rightarrow \infty} a_n = +\infty$, where $a_n = \operatorname{Re} z_n$; $|F(z_n)| = 2$, $n \in \mathbb{N}$. We apply Theorem 2.1 with δ equal to the chordal distance from $C(0; 2)$ to $\{0, 1, \infty\}$ and conclude that

$$\chi[F(z_n + 2\pi z), \infty] \geq m(\tfrac{1}{2}),$$

$|z| \leq \frac{1}{2}$, and hence that $|F(a_n + it)| \leq [m(\frac{1}{2})]^{-1}$, $t \in \mathbb{R}$, $n \in \mathbb{N}$. Since by the boundary maximum principle for analytic functions

$$|f(z)| \leq [m(\tfrac{1}{2})]^{-1},$$

$e^{a_n} \leq |z| \leq e^{a_{n+1}}$, $n \in \mathbb{N}$, f is bounded on $\{e^{a_0} \leq |z| < +\infty\}$ and hence has a removable singularity at ∞ .

EXERCISES

2.1. Given $f \in \mathbf{M}(\mathbb{C})$. Suppose that $\nu_r(w) < +\infty$ for three distinct w . Show that f is the restriction to $\mathbb{C}$ of a member of $\mathbf{M}(\hat{\mathbb{C}})$.

2.2. Show that the big Picard theorem as formulated is equivalent to: Let f be meromorphic on $\Delta(a; r) - \{a\}$. If f is not the restriction of a function meromorphic on $\Delta(a; r)$, then $f^{-1}(\{w\})$ clusters at a save for at most two values of w .